QUESTION BANK

**PCCST501 – COMPUTER NETWORKS
 MODULE 1**

Overview of the Internet • Protocol Layering • Application Layer

1. Syllabus and examination alignment

Module 1 carries 6 contact hours and covers: Overview of the Internet; Protocol layering; Application-layer paradigms; client–server applications (WWW and HTTP, FTP, Electronic Mail and DNS); peer-to-peer networks; and the BitTorrent case study.

Course Outcome alignment: CO1 – Understand the internetworking design in terms of protocol stack and the role of various application-layer protocols (K2).

The supplied model question paper uses the following ESE structure: Part A – answer all 8 questions, 3 marks each (24 marks); Part B – two 9-mark questions per module, answer one full question from each module; each full question may contain up to three subdivisions. The question bank below follows this style. Module 1 questions are mapped to CO1.

2. Part A – Short-answer questions (3 marks each)

AJK CSE

Answer each question briefly. Use a labelled diagram, example or comparison wherever relevant.

A. Overview of the Internet

1. Define a computer network. Distinguish between an end system (host) and a connecting device, with suitable examples.

2. Differentiate a LAN and a WAN based on geographical coverage and ownership. Give one example of each.

3. Explain the functions of a router, a switch and a modem in a computer network.

4. Distinguish between circuit switching and packet switching.

5. Explain the store-and-forward principle used in packet switching.

6. Differentiate between an Internet Service Provider (ISP) and an Internet Exchange Point (IXP).

7. List the main components required to access the Internet from a home or campus network and state the function of any three.

8. Differentiate between network hardware and network software with suitable examples.

B. Protocol layering and TCP/IP architecture

9. Define a protocol. State any three elements or rules that govern communication between network entities.

10. What is protocol layering? State two reasons for dividing network communication into layers.

11. Draw the TCP/IP protocol suite and name its layers.

12. State the major function of each layer in the TCP/IP protocol suite.

13. List the protocol data unit (PDU) associated with the application, transport, network and data-link layers.

14. Define encapsulation and decapsulation in the context of layered network communication.

15. Differentiate between logical communication and physical communication between two hosts.

16. What is a peer-layer protocol? Explain the term logical connection between corresponding layers.

17. Explain how the services of a lower layer are used by the layer immediately above it.

18. State two advantages of the layered architecture of the Internet.

C. Application-layer paradigms and sockets

19. What is meant by an application-layer protocol? State the role of an application-layer protocol in process communication.

20. Differentiate between the client–server and peer-to-peer paradigms. Give one example of each.

21. State three characteristics of a client–server application.

22. Define a socket. Why is a socket considered an endpoint of process-to-process communication?

23. What information is contained in a socket address? State the purpose of the IP address and port number.

24. Distinguish between a standard application-layer protocol and a nonstandard application-layer protocol.

D. WWW and HTTP

25. Define the World Wide Web. Mention two characteristics that contributed to its growth.

26. What is a URL? Identify the main components of a URL with an example.

27. Explain the request–response model of HTTP.

28. Differentiate between persistent and non-persistent HTTP connections.

29. Name any three HTTP request methods and state the purpose of each.

30. List the five broad categories of HTTP status codes and state the meaning of each category.

31. What is a cookie? Explain how a cookie helps a web server recognize a returning client.

32. Define a proxy server. State any two advantages of web caching.

E. FTP, Electronic Mail and DNS

33. Define FTP. Why does FTP use separate control and data connections?

34. State the functions and port numbers of the FTP control connection and data connection.

35. Differentiate between the User Agent (UA), Message Transfer Agent (MTA) and Message Access Agent (MAA) in e-mail architecture.

36. State the roles of SMTP, POP3 and IMAP in electronic mail.

37. What is MIME? Why is MIME required for electronic mail?

38. Define DNS and explain why a distributed naming system is required on the Internet.

39. Differentiate between recursive and iterative DNS resolution.

40. What is DNS caching? Explain the significance of the Time To Live (TTL) field.

41. Write the purpose of any three DNS resource record types: A, AAAA, NS, CNAME, MX and SOA.

F. Peer-to-peer networks and BitTorrent

42. Define a peer-to-peer network. State any three characteristics of the P2P paradigm.

43. Differentiate between centralized P2P and decentralized P2P networks.

44. What is an overlay network in the context of P2P systems?

45. Explain the terms torrent, swarm, seed and leech in BitTorrent.

46. What is the role of a tracker in a BitTorrent network?

47. Define choking and unchoking in BitTorrent.

48. What is optimistic unchoking? State its purpose.

49. Explain the rarest-first piece selection strategy used in BitTorrent.

50. What is a Distributed Hash Table (DHT)? How does it support trackerless BitTorrent?

3. Part B – Descriptive / analytical questions (9 marks each)

Each question is designed as a full 9-mark option. Subdivision marks are indicated. Draw neat, labelled diagrams wherever required.

A. Overview of the Internet and protocol layering

1. Explain the TCP/IP protocol suite with a neat layered diagram. Describe the function of each layer and identify the PDU handled at each layer. (5 marks) Trace the transmission of a message from an application on Host A to an application on Host B, illustrating encapsulation and decapsulation. Show the logical peer-layer communication. (4 marks)

2. With suitable diagrams, explain how a packet-switched network transfers data using the store-and-forward principle. Compare packet switching with circuit switching in terms of resource allocation, path establishment and sharing of links. (5+4 marks)

3. Explain protocol layering and the need for a layered architecture. Discuss the concepts of service, interface, protocol, peer entities and logical communication with a suitable example. (5+4 marks)

4. A message is sent from one host to another through intermediate routers. Explain the distinction between logical communication and actual physical communication. Illustrate the roles of end systems, routers and communication links in the journey. (4+5 marks)

B. Application-layer paradigms and WWW/HTTP

1. Explain the client–server paradigm with its characteristics and a neat architecture diagram. Compare it with the peer-to-peer paradigm using suitable examples. (5+4 marks)

2. Explain the architecture of the World Wide Web. Describe the components of a URL and the working of HTTP request and response messages with a

suitable example. (3+6 marks)

3. Explain the format of an HTTP request message and an HTTP response message. Include the request line, status line, header fields and message body with suitable examples. (5+4 marks)

4. Compare persistent and non-persistent HTTP connections. Explain HTTP cookies and their role in maintaining state in a stateless Web. (4+5 marks)

5. Explain the working of a proxy server in web caching. Illustrate cache-hit and cache-miss operations and discuss how caching reduces response time, traffic and server load. (6+3 marks)

C. FTP and Electronic Mail

1. With a neat diagram, explain the architecture of FTP. Describe the functions of the control connection and data connection, their port numbers and their lifetimes. (6+3 marks)

2. Explain the working of an FTP session from login to file transfer and termination. Include the purpose of USER, PASS, LIST, RETR, STOR and QUIT commands. (6+3 marks)

3. With a neat diagram, explain the architecture of electronic mail. Describe the roles of the User Agent (UA), Message Transfer Agent (MTA) and Message Access Agent (MAA). (6 marks)
Explain the roles of SMTP, POP3 and IMAP in sending and retrieving e-mail. (3 marks)

4. Trace the delivery of an e-mail from a sender to a receiver through the mail servers. Explain SMTP connection establishment, message transfer and connection termination. (6 marks)
Why is MIME required? Mention the types of content it enables e-mail to carry. (3 marks)

5. Compare POP3 and IMAP with respect to message storage, access from multiple devices, folders, synchronization and message retrieval. Explain why SMTP alone is insufficient for retrieving e-mail. (6+3 marks)

D. Domain Name System

1. Explain the hierarchical domain name space of DNS with a neat diagram. Describe the roles of root, top-level-domain and authoritative name servers. (6+3 marks)

2. Explain the steps involved in DNS name resolution when a user enters a domain name in a browser. Differentiate between recursive and iterative resolution with diagrams. (4+5 marks)

3. Explain DNS resource records and their format. Describe the purpose of A, AAAA, NS, CNAME, MX and SOA records with examples. (6+3 marks)

4. Discuss DNS caching and the role of TTL. Explain how DNS uses UDP and TCP and identify the well-known port number used by DNS. (5+4 marks)

E. P2P networks and BitTorrent case study

1. Explain the working of a peer-to-peer network. Compare centralized, decentralized-unstructured and decentralized-structured P2P networks with suitable examples. (5+4 marks)

2. Explain the architecture and working of a centralized P2P network using Napster as an example. Distinguish between directory lookup and the actual file-transfer operation. State its advantages and limitations. (6+3 marks)

3. Explain the working of an unstructured P2P network using Gnutella. Describe bootstrap, ping, pong and flooding. Discuss the scalability problem caused by flooding and the role of ultra nodes and query-routing improvements. (6+3 marks)

4. Explain the role of a Distributed Hash Table (DHT) in structured P2P networks. How does structured P2P improve file lookup efficiency and scalability compared with unstructured P2P? (5+4 marks)

5. With a neat diagram, explain the working of BitTorrent. Include the torrent file, tracker, swarm, seeds, leeches and chunk exchange. (6 marks)
Explain the rarest-first strategy and the purpose of choking, unchoking and optimistic unchoking. (3 marks)

6. Explain how a new peer joins a BitTorrent swarm and becomes a seed. Describe the role of the tracker and torrent metafile. (5 marks)
Explain trackerless BitTorrent and how DHT removes dependence on a central tracker. (4 marks)

4. Model-question-paper style practice sets

The following practice sets reproduce the Module 1 Part B pattern in the uploaded model paper: two alternatives, each carrying 9 marks.

Practice Set 1 – Module 1

Q1 (a) Explain the layered architecture of the TCP/IP protocol suite, stating the functions and PDUs of the layers. (5)
(b) Illustrate encapsulation and decapsulation when a message travels from Host A to Host B. (4)

OR

Q2 (a) Explain the architecture of electronic mail and the roles of UA, MTA and MAA. (6)
(b) Explain the purpose of the control and data connections in FTP. (3)

Practice Set 2 – Module 1

Q1 (a) Explain HTTP request and response message formats with suitable examples. (5)
(b) Explain how cookies and proxy caching support modern Web applications. (4)

OR

Q2 (a) Explain DNS hierarchy and the steps in recursive and iterative name resolution. (6)
(b) State the functions of any three DNS resource record types. (3)

Practice Set 3 – Module 1

Q1 (a) Compare centralized, unstructured and structured P2P networks. (5)
(b) Explain the role of DHT in structured P2P networks. (4)

OR

Q2 (a) Explain BitTorrent operation, including tracker, swarm, seed, leech and chunk exchange. (6)
(b) Explain rarest-first and optimistic unchoking. (3)